

Synthetic Data Generation for Economic Policymaking

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Introductions



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Learning objectives

1. classify a synthetic data design by what is observed, missing, and generated
2. write the synthetic data generation problem as $p(Y | X)$
3. generate multiple synthetic imputed datasets and analyze them
4. validate synthetic data with holdout data transportability checks
5. explain when synthetic data evidence is useful, limited, or not credible

Basic prerequisites

Minimum prerequisites includes:

- ▶ basic R literacy: run a script, edit a variable name, etc
- ▶ comfort with household survey concepts
- ▶ regression fundamentals: predictions depend on predictors
- ▶ willingness to critically assess model output

Let's get to know you

Answer these background questions so we all get to know each other.



<https://app.sli.do/event/aAtCCTfQy4TMpy51HkbBUf>

Datasets have their own strengths/weaknesses

- ▶ Have you ever tried to do a project and realize your data set is missing a **key variable**?
 - ▶ Moreover, that variable may be available in a different data set!
- ▶ **Can synthetic data help us combine complementary surveys for policy analysis?**
- ▶ We will introduce you to two surveys in this tutorial
 1. **GLSS** is strong on consumption-based welfare aggregates, poverty, and expenditure
 2. **DHS** is strong on health, fertility, nutrition, and household conditions.

Our two data sources: **GLSS** and **DHS**

	GLSS	DHS
What it is	Ghana Living Standards Survey	Demographic and Health Survey
Who runs it	Ghana Statistical Service	GSS, with health-sector partners and DHS Program support
Focus	Consumption, expenditure, income; official poverty statistics	Fertility, child health, nutrition, mortality, housing
Rounds	GLSS1 1987/88 to GLSS7 2016/17	1988 to 2022, roughly every five years

Examples of DHS 2014 variables

Role	DHS 2014 examples
Analysis outcomes	stunting/anthropometry, anemia, fever, diarrhea, cough, vaccination
Shared predictors X	region, urban/rural, household size, head age/sex, education, assets, electricity, water, toilet, cooking fuel, housing

- ▶ What the DHS 2014 lacks: consumption-based welfare aggregate, consumption, poverty status.
- ▶ We will call these “missing” variables the synthetic targets Y

Examples of GLSS6 variables

Role	GLSS6 examples
Synthetic targets Y	welfare aggregate, poverty status, total/food/non-food expenditure, health spending
Shared predictors X	region, urban/rural, household size, education, housing, water, toilet, electricity, cooking fuel

- ▶ **What the GLSS6 is missing:** child anthropometry outcomes (e.g., stunting), anemia, illness, and vaccination.
- ▶ We will call these the **analysis/policy outcomes**

GLSS6 vs DHS 2014

Welfare aggregate + expenditure

Health + nutrition

GLSS6
2012/13

impute welfare aggregate into DHS



DHS
2014

lacks DHS child outcomes

lacks GLSS poverty + consumption

Running question for today (need both surveys)

- ▶ **Running question: What share of children are stunted in poor vs. non-poor households, using Ghana's official consumption-based poverty line?**
- ▶ In **DHS**, stunted = 1 means the child's height-for-age z-score is below -2
 - ▶ In our **DHS** teaching extract, the weighted stunting rate is **17.9%**
 - ▶ **DHS** has stunting but no consumption-based poverty line
- ▶ **GLSS6** has the official poverty line but no stunting
- ▶ Synthetic data generation lets **DHS** borrow poverty status from **GLSS**

Prediction before we impute poverty

In our **DHS** teaching extract, the weighted stunting rate is **17.9%**.

**Once we impute consumption-based poverty,
should stunting be higher among poor households?**

Three variable groups drive the running question

Group	Variable in the tutorial	How to read it
Shared predictors X	region10, urban, household traits	observed in both surveys
Synthetic target Y	welfare aggregate (welfare)	real consumption/expenditure per adult equivalent in GLSS6 , imputed into DHS
Analysis outcome	stunted	a DHS child indicator based on height-for-age

The welfare aggregate is constructed

- ▶ **GLSS** asks many food and non-food consumption/expenditure questions.
- ▶ The GLSS poverty aggregate file combines those items into a real living standard measure.
- ▶ In this tutorial, welfare is **annual Ghana cedis per adult equivalent**, in 2013 prices.
- ▶ It is not household income, and it is not the **DHS** asset-based wealth index.

From the welfare aggregate to poverty

Poverty status is **not** imputed. It is a deterministic rule applied to each drawn consumption-based welfare aggregate (welfare):

$$\text{poverty status} = \begin{cases} \text{very poor} & \text{welfare agg.} < 792 \\ \text{poor} & 792 \leq \text{welfare agg.} < 1,314 \\ \text{non-poor} & \text{welfare agg.} \geq 1,314 \end{cases} \quad (\text{GH¢/adult eq.})$$

Think–pair–share: printed worksheet, Part A

- ▶ **Write one question** you could *only* answer by combining both surveys—name the **DHS** outcome and the **GLSS**-only variable it needs.
- ▶ Then, write the role for each variable in the blank column (on worksheet)

Role	What it means (not the answer)
Predictors X	observed in both surveys
Targets Y	observed in GLSS only; what we generate for DHS
Outcomes	DHS child outcomes studied after synthetic data generation

Another motivation: survey data between waves

- ▶ **GHAMOD** is Ghana's tax-benefit microsimulation model, part of UNU-WIDER's SOUTHMOD family.
- ▶ It applies Ghana's tax and benefit rules—income taxes, VAT, LEAP transfers—to **GLSS** household microdata, household by household, to evaluate reforms before they happen.
- ▶ Policy rules are updated every year, but recent **GLSS** rounds arrive only **every four to seven years** — so reforms get simulated on an aging population.
- ▶ Synthetic data is used to generate a current-year baseline between survey waves.

Synthetic data is a missing data framework

- ▶ We are in search of a model as follows:

$$\underbrace{Y^{mis} \sim p(Y \mid X, \theta)}_{\text{generate for DHS households}}, \quad \underbrace{\theta \sim p(\theta \mid Y^{obs}, X)}_{\text{learned entirely in GLSS}}$$

- ▶ $p(\cdot)$ means a **probability model**: it gives a distribution of possible welfare aggregate values for households with X .
- ▶ We learn the relationship from **GLSS**, where θ comes from
- ▶ We then **transport** it to **DHS**: plug each **DHS** household's X into the fitted model and draw its missing **synthetic targets** Y

We model columns in **GLSS**, then fill them in **DHS**

GLSS6 train

hh_id	region10	urban	hh_size	welfare agg. ('welfare')	stunted (outcome)
G001	Ashanti	1	5	1,120	--
G002	Northern	0	7	760	--

observed in GLSS

missing in DHS

fit p(Y | X) then fill column

DHS 2014 target

hh_id	region10	urban	hh_size	welfare agg. ('welfare')	stunted (outcome)
D001	Ashanti	1	4	?	0
D002	Northern	0	6	?	1

Step 1: Choose a data set to build the model on

- ▶ We build the model on a **GLSS6**: a harmonized extract of households carrying the **synthetic targets** Y plus the **shared predictors** X
- ▶ We need to recode the shared predictors to common categories
- ▶ **GLSS6** is the estimation sample (X and Y observed)
- ▶ **DHS 2014** supplies the records (households) to be completed (with only X)

Step 2: We create partially synthetic data

- ▶ We model only the imputed block, conditional on the shared predictors, as a chain of conditional marginals (sequential regression):

$$p(Y_1, \dots, Y_K \mid X) = p(Y_1 \mid X) p(Y_2 \mid X, Y_1) \cdots p(Y_K \mid X, Y_1, \dots, Y_{K-1})$$

- ▶ **Partially synthetic:** each **DHS** record keeps its real X ; only the **GLSS**-only **synthetic target** Y block is generated.
- ▶ Within the block, the chain models the joint distribution: each draw conditions on earlier draws
- ▶ See the appendix for fully synthetic data case.

Activity: Derive the model (worksheet Part B)

Your task: Derive the joint model of the two **synthetic targets** Y — **log welfare aggregate** ($\log(\text{welfare})$) and **NHIS registration** — into a chain of draws conditional on the **shared predictors** X .

$$p(\log(\text{welfare}), \text{NHIS} \mid X) = \underline{\hspace{2cm}}$$

Begin with single imputation

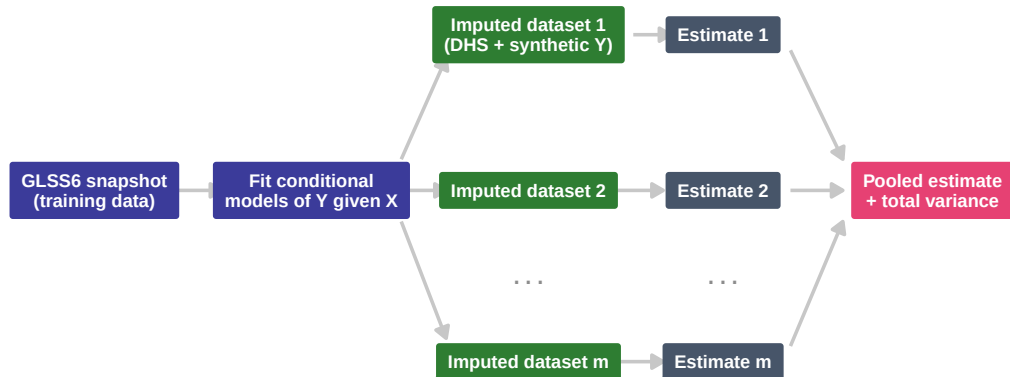
Start with a **single** imputation — one completed dataset:

1. Fit the model of Y on X in **GLSS** (e.g., a regression for log welfare aggregate).
2. **Draw the parameters** θ from their estimated distribution
3. **Draw a value** Y^{mis} for each **DHS** household from $p(Y | X, \theta)$.

That gives **one imputed dataset** (sometimes called an implicate): real X , synthetic Y .

But a single draw hides its own uncertainty — so we do not stop at one.

The imputation workflow at a glance



Same analysis on every imputed dataset; average estimates; total variance = within + between

Multiple imputation keeps uncertainty visible

- ▶ **Multiple imputation** (Rubin 1987): repeat the single-imputation draw several times, creating multiple completed copies of the data (**imputed datasets**), each with its own drawn values.
- ▶ Why not impute once? A single imputation hides the extra variability the imputation introduces — confidence intervals come out too small, and parameters look significant too often.
- ▶ Run the same standard analysis on each imputed dataset, then combine:
 - ▶ average the point estimates
 - ▶ $\text{total variance} = \text{within-imputation variance} + \text{between-imputation variance}$
- ▶ Never report results from a single imputed dataset.

Multiple imputation means multiple completed DHS copies

Only imputed cells change; observed DHS columns stay fixed

Imputed dataset 1	hh_id	region10	urban	hh_size	welfare agg. (imputed)	stunted (outcome)
	D001	Ashanti	1	4	1,050	0
	D002	Northern	0	6	820	1
Imputed dataset 2	hh_id	region10	urban	hh_size	welfare agg. (imputed)	stunted (outcome)
	D001	Ashanti	1	4	1,240	0
	D002	Northern	0	6	690	1
Imputed dataset 3	hh_id	region10	urban	hh_size	welfare agg. (imputed)	stunted (outcome)
	D001	Ashanti	1	4	1,180	0
	D002	Northern	0	6	940	1

Decisions for every imputed variable

1. **Model type** — parametric regression, logistic regression, CART, or hot-deck / predictive mean matching, etc.
2. **Restrictions on values** — keep expenditures non-negative; derive poverty status from the drawn welfare aggregate at the official poverty lines; structural zeros stay zero (e.g. education spending in a household with no one enrolled in school)

Two validation checks

We build in **two** ways to test our synthetic data model:

1. **GLSS holdout**: can the model recover the **synthetic targets** Y where we already know it?
2. **Transport check** — does a **GLSS** relationship still hold in **DHS**?

Validation Check 1 — GLSS holdout

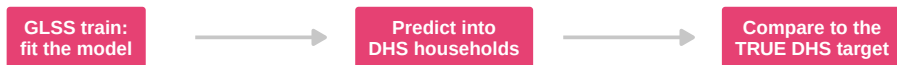
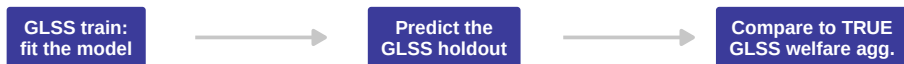
- ▶ We hold out a slice of **GLSS6** households **before** fitting.
- ▶ The **synthetic targets** Y (e.g., the welfare aggregate) is not observed in **DHS**
 - ▶ the holdout is the only place generated welfare aggregate values can be checked against ground truth.
- ▶ Fit on the rest, predict the held-out households, compare to their **true** welfare aggregate.

Validation Check 2 — transport

- ▶ Here we meet a fourth type of variable: a **validation target**: a **shared variable DHS** and **GLSS** *both* observe (e.g., electricity)
- ▶ Pretend **DHS** lacks it, predict it from **GLSS**, then compare to what **DHS** actually recorded.
- ▶ This tests whether the relationship hold across surveys (transport) for an observed variable
 - ▶ In the synthetic data literature this is called *calibration*

Summary of validation checks

Check 1, ground truth: recover the welfare aggregate where we KNOW it (held-out GLSS)



Check 2, transport: does a GLSS relationship hold in DHS (on a validation target)?

Judging usefulness of synthetic data

We will ask whether the synthetic data preserve:

- ▶ the welfare aggregate distribution in a **GLSS** holdout
- ▶ subgroup patterns across region and urban/rural status
- ▶ validation targets the model did not train on
- ▶ the final child-outcome gradient by synthetic poverty status

Pros and Cons of synthetic data

- ▶ Advantages of synthetic data
 - ▶ It can help analysts reason between survey waves.
 - ▶ It can support scenario analysis and model stress tests.
- ▶ Synthetic data can fail when:
 - ▶ the two surveys measure similar concepts differently
 - ▶ important variables are omitted
 - ▶ relationships differ across the **GLSS** and **DHS** samples

Transition to hands-on session

1. **Set up** — inspect the harmonized **GLSS**/**DHS** surfaces; define the welfare aggregate (continuous), derived poverty status, NHIS registration (binary), and the shared predictors.
2. **Split** — 80/20 region-stratified **GLSS** holdout, fixed seed.
3. **Fit** — the imputation model in **GLSS**, with household weights.
4. **Check twice** — welfare aggregate vs. ground truth on the holdout; one validation target into **DHS**.
5. **Enrich** — impute the welfare aggregate into **DHS**, $m = 5$ imputed datasets.
6. **Analyze and pool** — child outcomes by synthetic poverty status; Rubin within + between variance.

Setting up your computer

1. Download/clone the public GitHub repo and open that folder.
2. Install **R**, **RStudio**, and these R packages:

```
install.packages(c("mice", "dplyr", "ggplot2",  
                  "rmarkdown", "kableExtra", "tidyr"))
```

3. During the practical, download acet-indabax-2026-local-data.zip from the README Dropbox link.
4. Unzip it at the repo root so the files land in data/practical/teaching/.
5. Open the practical .Rmd notebook. Prefer line-by-line code? Open the generated .R script.
6. No local setup? Use the WebR HTML file in webr/.

Backup

Backup: Other use cases fit the same logic

- ▶ **Between-wave GLSS updating:** generate plausible policy-year baselines for GHAMOD-style microsimulation.
- ▶ **Crisis simulation:** stress-test social protection systems under income or price shocks.
- ▶ **Small-area analysis:** generate plausible local estimates when samples are thin.
- ▶ **Privacy-preserving access:** release synthetic microdata when raw records cannot be shared.

Backup: Partially synthetic and fully synthetic differ

Type	What is replaced?	Typical purpose
Partially synthetic	Sensitive variables or records	Reduce disclosure risk while preserving structure
Fully synthetic	All values are generated	Broader release or model-based simulation

Backup: The fully synthetic model

Fully synthetic data models the entire joint distribution — predictors included:

$$p(Y_1, \dots, Y_K) = p(Y_1) p(Y_2 \mid Y_1) \cdots p(Y_K \mid Y_1, \dots, Y_{K-1})$$

- ▶ Whole records are drawn from the model; no released value is an observed response (the privacy use case).
- ▶ Our enrichment needs only the conditional block $p(Y \mid X)$, because **DHS** supplies real X — that is what makes it partially synthetic.
- ▶ *One nuance:* classic partially synthetic data replaces observed

Backup: The SIPP Synthetic Beta is the canonical public example

- ▶ The Census Bureau created synthetic SIPP data linked to administrative earnings and benefits.
- ▶ Researchers can analyze synthetic files, then request validation against confidential “gold standard” data.
- ▶ The key lesson is institutional: synthetic data works best with a validation service and feedback loop.

Backup: Combining estimates across imputed datasets

For m imputed datasets with estimates $q^{(i)}$ and variances $u^{(i)}$,
Rubin's rules (what `mice` uses) give:

$$\bar{q} = \frac{1}{m} \sum_i q^{(i)}, \quad \bar{u} = \frac{1}{m} \sum_i u^{(i)}, \quad b = \frac{1}{m-1} \sum_i \left(q^{(i)} - \bar{q} \right)^2$$

$$T = \bar{u} + \left(1 + \frac{1}{m} \right) b$$

- ▶ $\bar{u}$ is the average within-imputation variance; b is the between-imputation variance.
- ▶ Fully synthetic replicates use a different total-variance formula (Reiter 2004), as in the SSB.

Backup: pMSE is a one-number utility check

- ▶ Stack real and synthetic records; create an indicator equal to 1 for synthetic rows.
- ▶ Fit a logistic regression of the indicator on the variables; pMSE is the mean squared deviation of the predicted probabilities from 0.5.
- ▶ If the model cannot separate real from synthetic, predictions sit near 0.5; the utility score $1 - \frac{pMSE}{0.25}$ runs from 0 (worst) to 1 (best).

Backup: The strongest critique is about joint distributions

Recent critiques of fully synthetic census microdata warn that realistic marginals do not guarantee realistic relationships.

Practical implication: our **DHS** enrichment validation should test the relationships that matter for policy, not just the variables that are easiest to plot.

Backup: References for deeper reading

- ▶ Rubin (1987): multiple imputation for nonresponse in surveys.
- ▶ Rubin (1993), Little (1993): synthetic data as multiple imputation.
- ▶ Benedetto, Stanley, and Totty (2018): SIPP Synthetic Beta.
- ▶ Jarmin, Louis, and Miranda: synthetic data at the U.S. Census Bureau.
- ▶ Freiman et al.: formal privacy and ACS synthetic data.
- ▶ Ruggles (2025): critique of fully synthetic census microdata.
- ▶ Adu-Ababio et al. (2023): GHAMOD v2.6 country report.